

9. Conditional statements

In order to perform conditional operations, java has introduced conditional statements. Some of the conditional statements are →

1. if condition
2. if-else condition
3. nested if condition
4. ladder if-else-if
5. Switch case

Let's have a look over it one by one-

1. If condition [🔗](#)

- 'if' conditional block comes with parenthesis and a body[curly braces]. where parenthesis is used to write a condition and body of block to implement the logic after condition is successfully validated.
- in parenthesis of if block boolean data type is expected as condition can be either true or false.
- Syntax for if condition->

```
1
2 if(condition){
3 //code to get executed if condition is 'true'
4 }
```

2. If-else condition [🔗](#)

- If condition can be attached with a else block. This block is used to write a logic which should be executed only if 'if' condition fails (condition is false in if , then else block will be executed).
- Syntax for if-else ->

```
1
2 if (condition){
3 // 'if' condition is true
4 }
5 else{
6 //'if' condition is false
7 }
```

3. nested-if condition [🔗](#)

- Consider there are two if conditions which should be inside other then such case is called as nested-if condition.
- E.g.- Consider from a table of Employees we want to find the females which are older than 21 years. So, in this case our first if will have condition to find all females out of Employee and another condition will find out females older than 21 years.
- Syntax with Example →

```
1 if(condition 1){
2 //condition 1 is to find female out of the Employee table
3
4     if(condition 2){
```

```

5 //condition 2 is to filter out females older than 21 years which are filtered by cond 1
6 }
7
8 }

```

4. *ladder if-else-if condition* [🔗](#)

- When there are more than 2 conditions then multiple blocks are needed where we can add conditions, in order to overcome this ladder if else if condition is introduced.
- Syntax for ladder if-else-if condition

```

1  if (color.equalsIgnoreCase("RED")) {
2      return "Stop";
3  }
4  else if (color.equalsIgnoreCase("YELLOW")) {
5      return "BE READY";
6  }
7  else if (color.equalsIgnoreCase("GREEN")) {
8      return "you can go";
9  }
10 else {
11     return "invalid input";
12 }

```

5. *Switch case condition* [🔗](#)

- Syntax for switch case

```

1 switch (expression) {
2     case value1:
3         // code to be executed if expression == value1
4         break;
5
6     case value2:
7         // code to be executed if expression == value2
8         break;
9
10    // more cases...
11    default:
12        // code to be executed if no cases match
13 }

```

- Example code for Switch case

```

1 import java.io.*;
2
3 class Geeks {
4     public static void main(String[] args)
5     {
6         int num = 20;
7         switch (num) {
8             case 5:
9                 System.out.println("It is 5");
10                break;
11            case 10:

```

```
12         System.out.println("It is 10");
13         break;
14     case 15:
15         System.out.println("It is 15");
16         break;
17     case 20:
18         System.out.println("It is 20");
19         break;
20     default:
21         System.out.println("Not present");
22     }
23 }
24 }
```

"If, at first, you do not succeed, call it version 1.0." — Khayri R.R. Woulfe

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